

Comparative Analysis of SRAM Cells in Sub-Threshold Region in 65nm

Raghav Madan
ECE, IIIT-Delhi
Delhi, India
raghav13134@iiitd.ac.in

Rishabh Gupta
ECE, IIIT-Delhi
Delhi, India
rishabh13137@iiitd.ac.in

Bhawana Singh Nirwan
ECE, IIIT-Delhi
Delhi, India
bhawana13152@iiitd.ac.in

Anuj Grover
STMicronics
India
anuj.grover@st.com

Abstract—Continuous technology scaling and the growing trend of low power applications have led to focus on ultra low voltage operating memories. This work presents a report on different configurations of SRAM cells (6T and 10T) to operate in sub-threshold voltage region.

Keywords—SRAM; sub-threshold; performance parameters; Eldo; Spice

I. INTRODUCTION

Power dissipation is of major concern in today's system-on-chip (SOC) designs with reducing technology nodes. With memories occupying the major chunk of area on a chip, it is of utter importance to also scale down the operating voltage of memories to reduce the overall power dissipation of a chip. To achieve the same, there is a compromise in terms of area or performance. Operating memories in sub-threshold voltage range increases write time and degrades performance in terms of static noise margin, read current, write noise margin, etc. as the technology shrinks.

This work presents a study of the conventional 6T SRAM memory cell and its inefficacy to give desired performance in the sub-threshold voltage range. Significant sub-threshold leakage is observed in the conventional 6T SRAM cell during standby mode. So it becomes difficult to retain data in the cell. Thus, the leakage power dominates in the overall power consumption as the SRAM cell remains in standby mode for most of the time. This leakage reduction becomes challenging in deep submicron technologies. The sub-threshold leakage and dynamic power consumption can be decreased if supply voltage is scaled down. However, this leads to an increase in circuit delay.

The paper is organized as follows. Section II describes the conventional 6T SRAM cell and its inefficiency in operating at sub-threshold voltage range. Section III includes the implementation of improved 6T and 10T cells to operate in sub-threshold range. Section IV presents the simulation results with conclusion in the next section. The simulations have been done on Spice as well as Cadence Virtuoso and verified using Eldo in 65nm technology.

II. 6T STANDARD CELL DESIGN

The 6T cell consists of a single word line and true and complementary bit lines. It contains a pair of cross coupled inverters and an access transistor for each bit line. Cell ratio (CR) and pull up ratio (PR) govern the proper sizing and the performance of the cell. These two terms are defined below:

$$CR = (W_3/L_3) / (W_6/L_6) \quad (1)$$

$$PR = (W_1/L_1) / (W_6/L_6) \quad (2)$$

Where, (W_i/L_i) is the width to length ratio of transistor M_i shown in Figure 1.

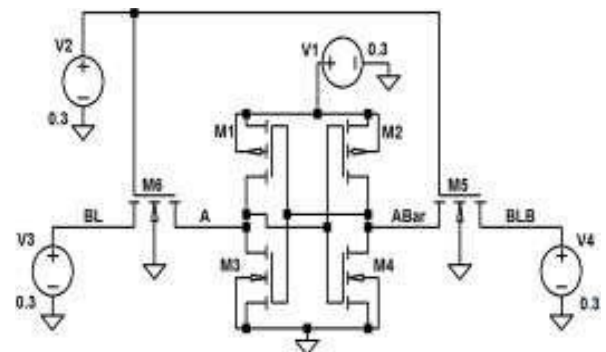


Fig. 1 Standard 6T SRAM cell

Voltage bump reduces with increased CR. If PR is increased, the cell voltage increases accordingly. In this work, the cell ratio was kept fixed and the performance of the cell with varying supply voltage ranging from 1.2V to 0.2V was studied. Degradation in cell performance was observed as the voltage was reduced down to the sub-threshold region. So, it is difficult to operate the conventional 6T SRAM cell in sub-threshold region. Results for different parametric observations are given in results section to verify the same. With these results, two improved configurations of 6T cell are studied. Also, a 10T SRAM cell configuration has been studied for performance comparison.

III. SRAM CELLS IN SUB-THRESHOLD

The design of the SRAM cell proposed in [1] is a 6T SRAM cell which is based on 4T PLNA (PMOS Latch, NMOS Access) cell. The cell operates in sub-threshold region as it overcomes the issues of leakage power during standby mode.

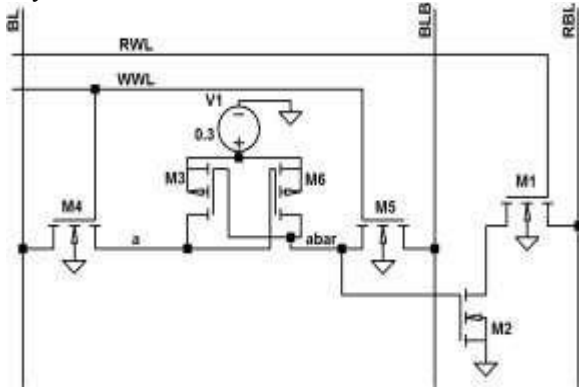


Fig. 2 4T PLNA based 6T SRAM cell

The cell schematic is shown in Figure 2. The concept of negative voltage on a word line is introduced which brings the transistor of read circuitry in super cut-off region during standby mode. Hence leakage is reduced. Performance of SRAM cell is improved as the read circuitry is isolated from write circuitry which eliminates the race condition during write as in the case of standard 6T SRAM cell. The circuit is also free from read static noise margin (SNM). This elimination of race condition saves significant dynamic power during data flipping.

The write and read operation of the cell somehow describe its performance. The write operation is done the same way as that of standard 6T SRAM cell. It starts with asserting the write word line (WWL) while holding either one of Bit Line (BL) or Bit line Bar (BLB) at V_{dd} and the other one at 0. The sizes of PMOS and NMOS in write circuitry is kept minimal. The size of access and pull-up transistors should be comparable to each other to retain the data. During read operation, read bit line (RBL) is pre-charged to V_{dd} and the remaining operation is self explanatory.

During standby mode if '1' is stored in the cell then both transistors of read circuitry are off and hence leakage will be very less due to stacking effect. But if '0' is stored, then the lower NMOS of read circuitry will be on, leading to higher leakage. To overcome this, negative voltage on read word line is used during standby that brings the upper NMOS of read circuitry in super cut-off region. The negative voltage is not given to write word line during standby mode as the PMOS in write circuitry leaks more than NMOS which can flip the data. To avoid this same amount of leakage is required to happen in NMOS of

PLNA in order to cancel out the effect of PMOS leakage. Hence data can be retained.

Another design, dynamic threshold based MOS (DTMOS) 6T SRAM cell design was proposed in [2]. The cell schematic is shown in Figure 3. This cell works well in the voltage range of 0.3- 0.4V. The read and write operations are done in the same way as in the standard 6T cell. However, the inverters used in the latch of SRAM cell are DTMOS inverters in which the gate and body of the transistors are tied together. This causes a body effect due to which the threshold voltage of the transistor changes dynamically based on the input to the device. This leads to improved performance of the cell as explained further.

During OFF state of the device, threshold voltage V_{th} increases reducing leakage from the device. Switching speed (low to high) of the inverters increases in active mode. However static leakage current reduces in standby mode due to increased V_{th} of the NMOS device. Hence, data is retained for a longer time with less leakage. Optimum sizing of the transistors of the cell is done to achieve desired performance.

The next architecture studied is of 10T SRAM cell as proposed in [3]. It works efficiently in sub-threshold region around 0.3V by separating read and write operations. This 10T bit cell provides improved noise margin by having separate read and write circuitry achieving high cell stability. A technique called dynamic differential cascade voltage switch logic (DCVSL) is used for reading mechanism which helps in compensating bit line leakage noise and hence increasing bit line swing.

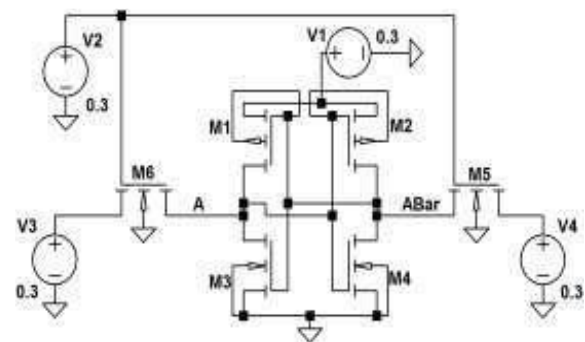


Fig. 3 DTMOS based 6T SRAM cell

It consists of two word lines 'W' (read) and 'WL' (write) used to decouple the read operation from write operation. In the read mode, 'W' line is enabled and 'WL' line is disabled through which it is decouples the data stored at nodes 'A' and 'ABar' from the bit lines. Due to this, its read SNM is improved, improving the read stability. In this cell, one of the two precharged bit lines discharges during read operation through the inverted lower NMOS device. That's why the position of bit line and inverted bit line are interchanged in this cell.

During write mode, both “W” (read) and “WL” (write) word lines are enabled with some boost voltage higher than V_{dd} to maintain good write ability as in sub-threshold region write ability is a critical issue. However in our work, we achieved the similar performance by keeping “WL” (write) line at V_{dd} and boosting the other word line higher than V_{dd} . There are two access transistors on each side which also affect write ability and both should be ON to successfully transfer the data from bit line to node “A”. By giving boost voltages on word lines, more write margin and less write time are achieved without area overhead penalty. To save more area, ground lines of all transistors are connected to common V_{gnd} .

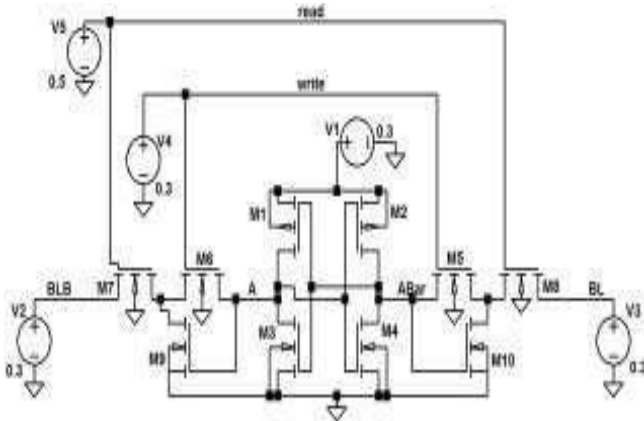


Fig. 4 10T SRAM cell

IV. SIMULATION RESULTS

The design of standard 6T SRAM cell is done in Spice as well as in Cadence Virtuoso at 65nm technology node and the simulation is done in Eldo. The supply voltage is varied from standard 1.2V to sub-threshold 0.2V and various parameters such as static voltage noise margin (SVNM), static current noise margin (SINM), write time, read current, leakage are evaluated. The results for all these parameters can be seen graphically in figures below. It is apparent that the standard 6T SRAM cell does not work satisfactorily in sub-threshold region.

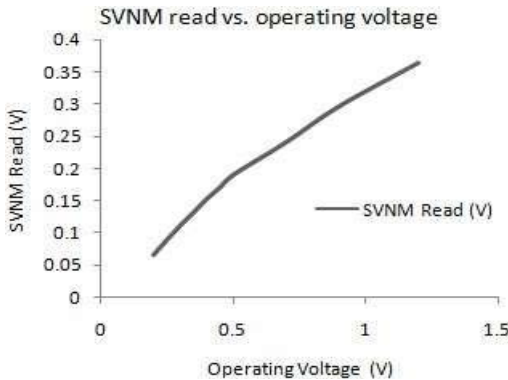


Fig. 5(a) Read SVNM varying with operating voltage

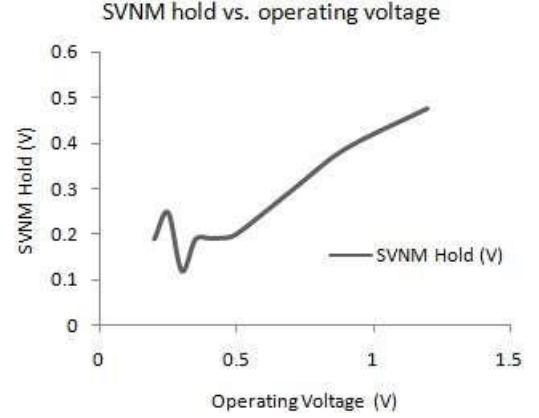


Fig. 5(b) Hold SVNM varying with operating voltage

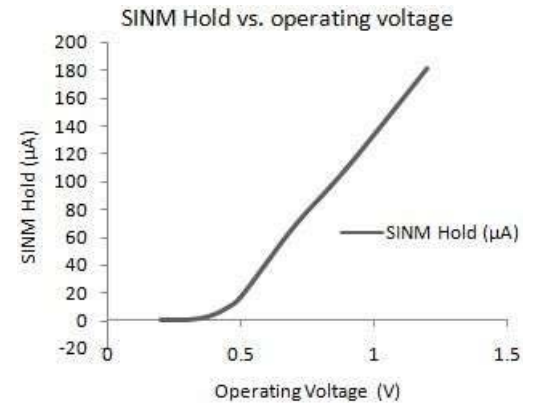


Fig. 6(a) Hold SINM varying with operating voltage

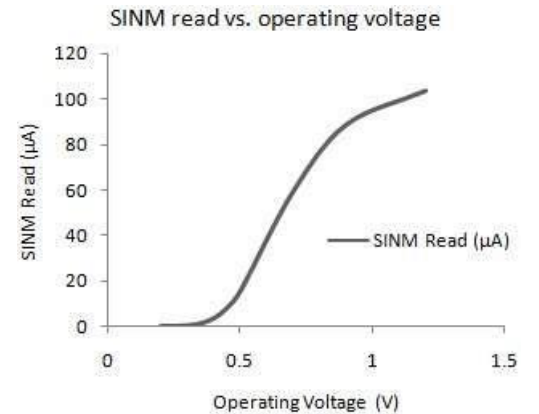


Fig. 6(b) Read SINM varying with operating voltage

It can be observed in Figure 5(a), (b) and Figure 6(a), (b) that as the supply voltage decreases from 1.2V to 0.2V, the SVNM and SINM also decreases. Hence the cell becomes more prone to noise and becomes less stable. In this work, N-curve method is used to find the SVNM and SINM for read stability rather than evaluating SNM using butterfly curve method [4]. Butterfly curve method is not used due to the fact

that after measuring the butterfly curves of the cell the SINM still needs to be derived by mathematical manipulation of the measured data. So the N-curve of the cell is used, which is measurable by inline testers and provides information about both voltage and current for read stability. By using the combined SVN and SINM, the read stability criteria for the cell are defined properly.

Further, write time and read current for the same configuration are also studied. A continuous degradation in the performance of the cell is observed as the supply voltage is reduced. Figure 7(a) and (b), depicts that as the supply voltage is reduced, the write time increases and the read current decreases which shows that read and write operations are affected by scaling down the supply voltage. However, the cell leakage decreases with supply voltage. Fig 7(c) depicts that the reduction in leakage of standard 6T SRAM cell when the voltage is decreased from 1.2V to 0.2V

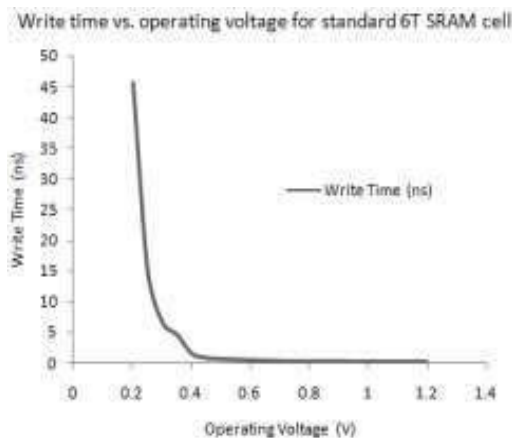


Fig. 7(a) Variation of write time with operating voltage

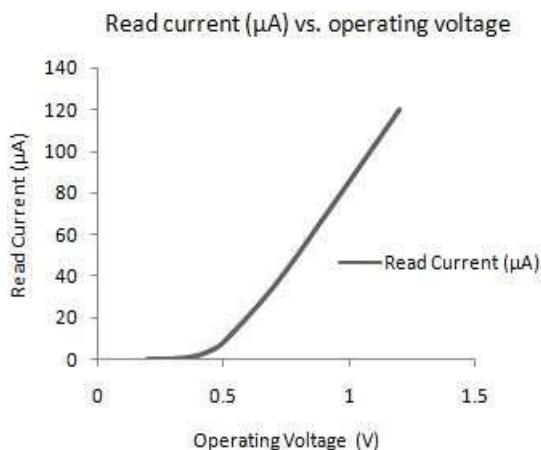


Fig. 7(b) Variation of read current with operating voltage

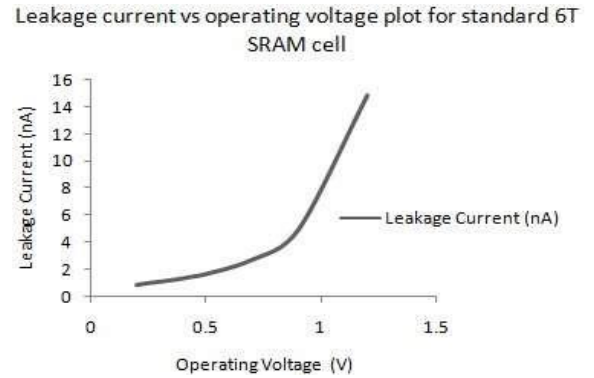


Fig. 7(c) Variation of leakage current with operating voltage

In order to make SRAM cell to operate in sub-threshold region the previously described cells are studied and simulated. These designs of SRAM cells work perfectly in sub-threshold region as inferred from the results. The sizing of transistors is done to obtain optimized results.

Table I summarizes the comparison of design parameters in terms of SNM, write time, read current, aspect ratio and cell area at $V_{dd} = 0.3V$.

TABLE I
PERFORMANCE COMPARISON BETWEEN STUDIED DESIGNS AT SUB-THRESHOLD VOLTAGE, $V_{DD} = 0.3V$

Parameter	Cell Type		
	4T PLNA based 6T	DTMOS based 6T	10T
Cell leakage current (nA)	1.220	2.136	2.019
Write time (ns)	0.5	2.7	2.4
SNM Hold (V)	0.225	0.121	0.127
SNM Read (V)	Read SNM free	0.118	Read SNM free
WNM (V)	0.1	0.055	0.080

It can be inferred that the 4T PLNA based 6T SRAM cell performs best in sub-threshold region with minimum leakage current, minimum write time and maximum hold SNM among the three designs. Also, this design is read SNM free.

V. CONCLUSION

In this paper a study of conventional 6T SRAM cell and its inability to operate in sub-threshold voltage region are discussed. It is verified that the performance degrades with reducing supply voltage. Further three SRAM cell configurations are implemented to operate successfully in sub-threshold region. The SNM calculation is done using the N-curve method by calculating SVN and SINM of the cells. All the designs are implemented at 65nm technology node.

REFERENCES

- [1] Yinhui Chen, Zhiyuan Yu, Haiqing Nan and Ken Choi, "Ultralow Power SRAM Design in Near Threshold Region using 45nm CMOS Technology," *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955. (*references*)
- [2] Myeong-Eun Hwang and Kaushik Roy, "A 135mV 0.13 μ W Process Tolerant 6T Sub-Threshold DT MOS SRAM in 90nm Technology" Custom Integrated Circuits Conference, IEEE, 2008.
- [3] Ik Joon Chang, Jae-Joon Kim, Sang Phill Park and Kaushik Roy, "A 32 kb 10T Sub-Threshold SRAM Array With Bit-Interleaving and Differential Read Scheme in 90 nm CMOS," *IEEE Journal of Solid-State Circuits*, vol. 44, no. 2, Feb. 2009
- [4] Evelyn Grossar, Michele Stucchi, Karen Maex and Wim Dehaene, "Read Stability and Write-Ability Analysis of SRAM Cells for Nanometer Technologies," *IEEE Journal of Solid-State Circuits*, vol. 41, no. 11, Nov. 2006